synthesis of essential components of DNA/RNA which increase rapidly during growth.

Problems during Pregnancy

- Nausea and vomiting
- Constipation
- Heart burn
- Oedema and leg cramps
- Pica
- Anemia
- Pregnancy induced hypertension (PIH)
- Gestational diabetes

Lactation

Adequate nutrition for the mother during lactation is also of vital importance as the infant is dependent on mother's milk for its nutrition for the first few months of life. Inadequate nutrition during lactation is reflected on both the quality and quantity of milk secreted.

Physiology of lactation

The Table 13 below gives the summary of hormonal control of lactation.

Nutrient Needs during Lactation

ICMR nutrient recommendations for a lactating mother are based on the composition of breast milk and the fact that 850 ml of milk is produced daily. However, the milk secretion continues to increase in the early periods of lactation up to six months and then gradually decreases. Therefore the nutrient requirements are given for the two periods in lactation i.e. 0-6 months and 6-12 months.

Energy

The lactating mother requires additional energy for the production of milk. Based on the optimal output of 850 ml/day, the additional allowance is recommended during first six months of lactation.

Protein

Due to production of milk, protein requirement also increases.

Fat

The total fat in breast milk is not influenced by the mother's diet. The fat also provides energy density to meet the higher energy requirement during lactation.

Calcium

The requirement for calcium doubles during lactation.

Iron

Since most mothers have lactational amenorrhea, it results in saving of nearly 1mg iron per day which otherwise would

Table 13 Summary of Hormonal Control of Lactation

S. No.	Hormone	Source	Function
1	Estrogen	Ovary	Stimulates breast development
2	Progesterone	Placenta	Prepares breast for milk production by changing glandular cells to secreting cells
3	Prolactin	Anterior pituitary	Stimulates milk production
4	Oxytocin	Posterior pituitary	Facilitates release of milk from alveolus

Source: Srilakshmi (2011)

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